

Monetary Economics and Policy

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Chapter 12: The Inflation
Unemployment Trade Off



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What We Will Learn

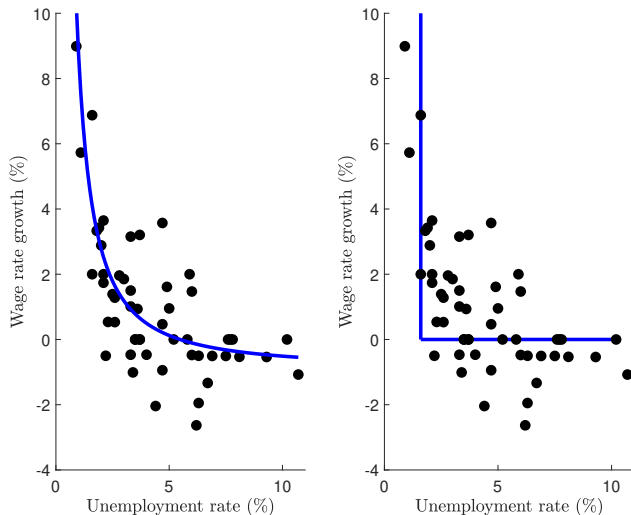
- ▶ Phillips (1958): evidence on the **nonlinear** relationship between wage inflation and unemployment.
- ▶ Why the **classical** view rejects a long-run trade-off: the economy returns to a **natural rate** once expectations adjust.
- ▶ How the Friedman–Phelps logic delivers an **expectations-augmented** Phillips curve: only **unexpected** inflation moves real activity in the short run.
- ▶ How New-Keynesian pricing frictions generate the **NKPC** and **persistent** monetary non-neutralities.
- ▶ How downward nominal wage rigidity and capacity constraints can imply an **L-shaped** Phillips curve:
 - ▶ flat in slack labor markets (“missing disinflation”),
 - ▶ steep near full employment (post-COVID inflation).

Outline

- 1 Introduction
- 2 Outline of the Results
- 3 Natural Rate of Unemployment
- 4 Keynesian Unemployment
- 5 Short-Run Non-Neutralities
- 6 Persistent Monetary Non-Neutralities
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Phillips (1958): Wage Inflation vs. Unemployment



Left panel: Phillips (1958). Right panel: illustrative L-curve.

Phillips (1958): Key Empirical Findings

- ▶ Phillips (1958) studies U.K. data (1861–1957) using scatter plots of:
 - ▶ wage inflation (rate of change of wage rates),
 - ▶ the unemployment rate.
- ▶ Core finding for the pre-WWI sample (1861–1913):
 - ▶ a strong **negative** relationship between wage inflation and unemployment,
 - ▶ a clear **non-linearity**: wage inflation rises much faster when unemployment is reduced from already low levels.

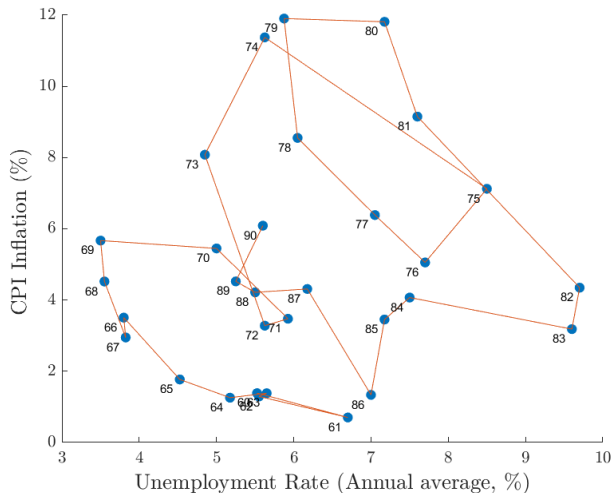
“An Empirical Finding in Search of Theory”

- ▶ Tobin (1972) famously described the Phillips curve as “an empirical finding in search of theory.”
- ▶ Phillips’ narrative:
 - ▶ low unemployment leads firms to compete for scarce workers, bidding wages up;
 - ▶ high unemployment makes workers reluctant to accept wages below prevailing rates.
- ▶ Keynes challenged the orthodox Great Depression interpretation:
 - ▶ observed wage stickiness does not imply voluntary unemployment;
 - ▶ workers accept lower **real** wages via higher prices more readily than nominal wage cuts;
 - ▶ nominal wages behave as “administered prices” (posted and fixed for periods);
 - ▶ workers care about their **relative** wage across firms.

From Phillips to the Natural Rate of Unemployment

- ▶ A literal Phillips reading suggests no unique “full-employment” anchor—rather, a menu of inflation/unemployment pairs.
- ▶ The immediate classical objection: wage/price inflation cannot permanently affect unemployment.
- ▶ Friedman (1969) and Phelps (1968):
 - ▶ confine the inflation–unemployment trade-off to the **short run**;
 - ▶ introduce **inflation expectations**;
 - ▶ define a **natural rate of unemployment**, determined by labor-market fundamentals and invariant to monetary policy in the long run.
- ▶ In the 1970s, adverse supply shocks (e.g. energy/oil) shifted the curve and complicated identification.

Great Inflation: Inflation vs. Unemployment



NK Revival and the Return of Capacity Constraints

- ▶ In the 1990s, New-Keynesian models micro-founded a Phillips curve through price-setting frictions:
 - ▶ staggered pricing implies forward-looking decisions;
 - ▶ monetary shocks can have more **persistent** real effects.
- ▶ But the debate shifted away from Keynes' original emphasis:
 - ▶ Keynes did not reject full employment; he recast it as a **capacity ceiling** (maximum aggregate supply);
 - ▶ an "L-shaped" view: in slack times, demand raises activity with little inflation, whereas in tight markets demand becomes inflationary.
- ▶ Connection to Friedman's **plucking model**: expansions rarely exceed the ceiling, while contractions "pluck" output below it.

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Outline of the Chapter

- ▶ Derive the **natural rate of unemployment** and show why it is independent of monetary policy in the long run.
- ▶ Introduce **downward nominal wage rigidity**: in slack markets unemployment becomes demand-determined, while tight markets revert to the natural-rate outcome.
- ▶ Show how information frictions (delayed information / preset wages) generate a **short-run** Phillips curve in the spirit of Friedman–Phelps.
- ▶ Present the **New-Keynesian Phillips curve (NKPC)** and explain why forward-looking price setting implies **persistent** monetary non-neutralities.
- ▶ Combine wage rigidity with the NKPC to obtain an **L-shaped New-Keynesian Phillips curve** that reconciles flat-slack and steep-tight regimes.

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Households: Heterogeneous Members and Labor Disutility

- ▶ We adopt a model of the labor market similar to Galí (2009).
- ▶ Large number of identical households.
- ▶ Each household has a continuum of members indexed by $(i, z) \in [0, 1] \times [0, 1]$:
 - ▶ i : labor type (or age/cohort),
 - ▶ z : disutility parameter; if the household is working, disutility is z^η with $\eta \geq 0$, otherwise 0.

Per-period utility:

$$U_t = \ln C_t - \int_0^1 \left(\int_0^{L_t(i)} z^\eta dz \right) di = \ln C_t - \int_0^1 \frac{(L_t(i))^{1+\eta}}{1+\eta} di,$$

where $L_t(i) \in [0, 1]$ is the employment fraction of type i .

Intertemporal Utility and Consumption Aggregator

Households maximize:

$$E_{t_0} \left\{ \sum_{t=t_0}^{\infty} \beta^{t-t_0} \left[\ln C_t - \int_0^1 \frac{(L_t(i))^{1+\eta}}{1+\eta} di \right] \right\}, \quad (12.1)$$

with rate of time preferences $0 < \beta < 1$.

- ▶ Consumption is a Dixit–Stiglitz aggregator of a continuum of consumption goods $C_t = \left[\int_0^1 C_t(j)^{\frac{\theta-1}{\theta}} dj \right]^{\frac{\theta}{\theta-1}}$ where $\theta > 1$ is the elasticity of substitution between goods $C(j)$ with $j \in [0, 1]$.
- ▶ Households optimize with respect to their consumption, saving plans. We assume they have access to an ample spectrum of assets to trade in and that financial markets are complete.

Flow Budget Constraint

Flow budget constraint:

$$B_t + P_t C_t = \mathcal{W}_t + \int_0^1 W_t(i) L_t(i) di + \Psi_t + \Psi_t^w - T_t. \quad (12.2)$$

- ▶ B_t is the value of a portfolio with next-period state-contingent payoff $\mathcal{W}_{t+1}$ such that $B_t = E_t(\tilde{R}_{t,t+1} \mathcal{W}_{t+1})$ and nominal discount factor $\tilde{R}_{t,t+1}$ defined in Chapter 5.
- ▶ Right-hand side represents resources held a time t : The payoff of their portfolio $\mathcal{W}_{t+1}$, income $W_t(i)L_t(i)$ for each cohort of workers i and nominal wage $W(i)$, firm profits Ψ less taxes T and profits of the workers' unions Ψ^w .
- ▶ Left-hand side represents spending: Households purchase consumption goods C_t at price P_t and the portfolio of value B_t .
- ▶ The household problem yields a standard intertemporal budget constraint as in Chapter 5.

Euler Equation

First-order conditions:

$$\frac{1}{P_t C_t} = \frac{\beta}{\tilde{R}_{t,t+1}} \frac{1}{P_{t+1} C_{t+1}}.$$

- ▶ The above equation characterizes optimal consumption/saving choices at each point in time t .
- ▶ Looking forward from time t , there is a set of first-order conditions (Euler equations) equal to the number of states of nature at time $t + 1$.
- ▶ Optimality implies that the intertemporal budget constraint of the consumers holds with equality.

Firms: Technology

- ▶ Continuum of firms $j \in [0, 1]$ with decreasing returns to labor:

$$Y_t(j) = A_t (L_t(j))^{\alpha_L}, \quad 0 < \alpha_L \leq 1,$$

where A_t denotes technology.

- ▶ Each firm hires a composite labor input $L_t(j)$ built from differentiated labor types $i \in [0, 1]$.

Firms: Labor Aggregation and Wage Index

- Firm-level labor aggregator across types i with elasticity $\theta_w > 1$:

$$L_t(j) = \left(\int_0^1 (L_t(i))^{\frac{\theta_w-1}{\theta_w}} di \right)^{\frac{\theta_w}{\theta_w-1}}.$$

- Given nominal wages $W_t(i)$, define the aggregate wage index:

$$W_t = \left(\int_0^1 (W_t(i))^{1-\theta_w} di \right)^{\frac{1}{1-\theta_w}}.$$

Firms: Demand and Profits

- Demand for firm j (as in Chapter 5) with $\theta > 1$:

$$Y_t(j) = \left(\frac{P_t(j)}{P_t} \right)^{-\theta} Y_t.$$

- Profits (with revenue tax τ_t):

$$\Psi_t(j) = (1 - \tau_t) P_t(j) Y_t(j) - W_t L_t(j).$$

- Intuition: profits rise with demand $Y_t(j)$ and fall with the wage bill $W_t L_t(j)$.

Price Setting and Labor Demand

- ▶ As in Chapter 5, optimal price setting implies a constant markup over marginal cost:

$$\mu_t \equiv \frac{\theta}{(\theta - 1)(1 - \tau_t)\alpha_L}, \quad P_t(j) = \mu_t \frac{W_t}{A_t} (L_t(j))^{1-\alpha_L}.$$

- ▶ In a symmetric equilibrium, $P_t(j) = P_t$ and $L_t(j) = L_t$.

Aggregate labor demand:

$$L_t = \left(\frac{\mu_t}{A_t} \frac{W_t}{P_t} \right)^{\frac{1}{\alpha_L - 1}}. \quad (12.3)$$

- ▶ Labor demand is decreasing in the real wage W_t/P_t since $\alpha_L - 1 < 0$.

Labor demand for type i :

$$L_t(i) = \left(\frac{W_t(i)}{W_t} \right)^{-\theta_w} L_t. \quad (12.4)$$

Unions: Objective and Optimal Wage Setting

- Union for type i chooses $W_t(i)$ taking demand (12.4) as given.

Objective function:

$$\Psi_t^w = \lambda_t W_t(i) L_t(i) - \frac{L_t(i)^{1+\eta}}{1+\eta}, \quad (12.5)$$

where $\lambda_t = \frac{1}{P_t C_t}$ converts nominal income into marginal utility units.

First-order condition (monopoly wage setting):

$$\frac{W_t(i)}{P_t} = \mu^w C_t L_t(i)^\eta.$$

with markup $\mu^w \equiv \frac{\theta_w}{\theta_w - 1}$.

Natural Employment Rate

- In a symmetric equilibrium, $W_t(i) = W_t$ and $L_t(i) = L_t$ for all i .

This implies the **labor-supply** relation:

$$L_t = \left(\frac{1}{\mu^w} \frac{W_t}{P_t C_t} \right)^{\frac{1}{\eta}}. \quad (12.6)$$

The intersection between labor demand (12.3) and labor supply (12.6) yields the **natural rate of employment**:

$$L_t^n = \left(\frac{1}{\mu^w} \frac{1}{\mu_t} \right)^{\frac{1}{1+\eta}}.$$

- In the flexible-wage equilibrium, the real wage adjusts so that employment equals L_t^n .
- Natural employment L_t^n depends only on goods-market and labor-market markups (μ_t, μ^w) .
- It does not depend on demand conditions or inflation.

Unemployment Rate (following Galí (2009))

- ▶ A type- i worker supplies labor iff the real wage covers disutility:

$$\frac{W_t(i)}{P_t} \geq C_t z^\eta.$$

- ▶ The marginal worker defines labor-force participation $L_t^s(i)$:

$$\frac{W_t(i)}{P_t} = C_t (L_t^s(i))^\eta.$$

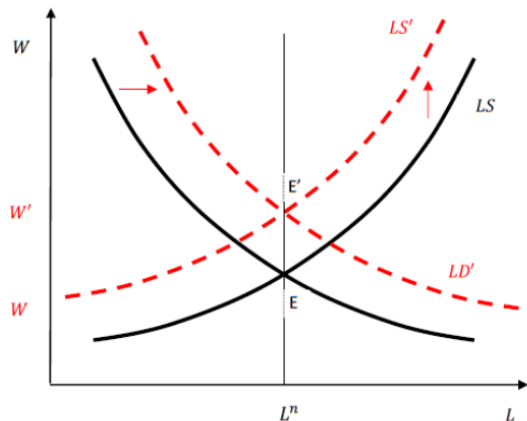
- ▶ In symmetry, $W_t(i) = W_t$ and $L_t^s(i) = L_t^s$, so

$$L_t^s = \left(\frac{W_t}{P_t C_t} \right)^{\frac{1}{\eta}}. \quad (12.7)$$

- ▶ Comparing (12.7) with (12.6) yields the **natural unemployment rate**:

$$u^n \equiv 1 - \frac{L_t^n}{L_t^s} = 1 - \left(\frac{1}{\mu^w} \right)^{\frac{1}{\eta}}. \quad (12.8)$$

Equilibrium Employment



An increase in the price level P_t shifts both LD and LS upward to LD' and LS' , raising the nominal wage while leaving employment at L^n .

- ▶ Labor supply and demand are plotted in **nominal wage–employment** space.
- ▶ A proportional increase in P_t shifts both LS and LD upward by the same factor.
- ▶ **Result:** employment remains at the natural level L^n ; only the nominal wage scales up.
- ▶ Hence, with flexible prices and wages, unemployment is **independent** of inflation.

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Keynesian Unemployment via Downward Nominal Wage Rigidity

- ▶ Assume nominal wages cannot fall below a lower bound $\bar{W}$.
- ▶ Unions again maximize (12.5) under demand (12.4) but with the constraint

$$W_t(i) = \max \{ \bar{W}, \mu^w C_t L_t(i)^\eta \}.$$

- ▶ This again implies symmetry across different cohorts $W_t(i) = W_t$ and $L_t(i) = L_t$.
- ▶ Implications for labor supply:
 - ▶ If the labor market is slack: $W_t = \bar{W}$ binds and labor supply is perfectly elastic at $\bar{W}$.
 - ▶ If the labor market is tight: $W_t > \bar{W}$ implying a return to the flexible-wage upward-sloping schedule.

Labor Demand in Terms of Nominal Spending

- By using $Y_t = A_t L_t^{\alpha_L}$ and defining nominal spending as $\tilde{Y} = P_t Y_t$ we can rewrite labor demand (12.3):

$$L_t = \frac{1}{W_t} \frac{\tilde{Y}_t}{\mu_t}.$$

- When the wage floor binds ($W_t = \textcolor{red}{W}$), employment becomes:

$$L_t = \frac{1}{\textcolor{red}{W}} \frac{\tilde{Y}_t}{\mu_t}.$$

- This shows that higher nominal spending can directly boost employment in slack markets.

Unemployment at the Wage Floor

- ▶ When $W_t = \textcolor{red}{W}$ and using $P_t C_t = P_t Y_t = \tilde{Y}_t$, labor force is given by:

$$L_t^s = \left(\frac{\textcolor{red}{W}}{\tilde{Y}_t} \right)^{\frac{1}{\eta}}$$

- ▶ Recall that employment is $L_t = \frac{1}{\textcolor{red}{W}} \frac{\tilde{Y}}{\mu_t}$, unemployment rate is then given by:

$$u_t = 1 - \frac{L_t}{L_t^s} = 1 - \frac{1}{\mu_t} \left(\frac{\tilde{Y}_t}{\textcolor{red}{W}} \right)^{1 + \frac{1}{\eta}}.$$

- ▶ u_t decreases in nominal spending $\tilde{Y}_t$ and increases in the markup μ_t .
- ▶ Policy that increases nominal spending $\tilde{Y}_t$ reduced unemployment, when labor market is slack.

Threshold Nominal Spending

- Define $\tilde{Y}_t^*$ as the nominal spending level at which the flexible wage exceeds the norm $\mathbf{W}$. The threshold is:

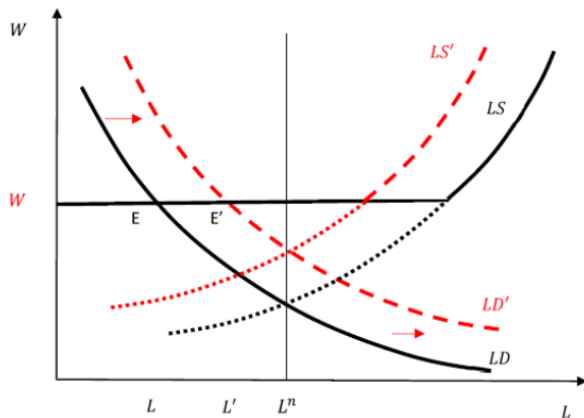
$$\tilde{Y}_t^* = \frac{\mathbf{W}}{\mu_t} \left(\frac{1}{\mu^w \mu_t} \right)^{\frac{\eta}{1+\eta}}.$$

- If $\tilde{Y}_t < \tilde{Y}_t^*$: wage floor binds, level of unemployment is driven by insufficient aggregate demand.
- If $\tilde{Y}_t \geq \tilde{Y}_t^*$: $W_t > \mathbf{W}$, economy returns to the flexible-wage outcome:

$$L_t = L_t^n = \left(\frac{1}{\mu^w \mu_t} \right)^{\frac{1}{1+\eta}},$$

with unemployment at the natural rate (12.8).

Equilibrium Employment



- ▶ Wage floor: $W_t = W$. If $\tilde{Y}_t < \tilde{Y}_t^*$, the constraint binds and employment is demand-determined.
- ▶ Initial equilibrium E : $L < L^n$ and $u > u^n$.
- ▶ Higher nominal spending shifts $LD \rightarrow LD'$ (and $LS \rightarrow LS'$), moving the economy to E' .
- ▶ Result: $L \uparrow$ and $u \downarrow$. Nominal spending reduces unemployment.

Asymmetry and the Plucking Interpretation

- ▶ **Keynesian asymmetry:**
 - ▶ **Slack is common:** when unemployment is above its natural rate, higher aggregate demand can raise L_t with limited inflation pressure.
 - ▶ **Tightness is exceptional:** near full employment, capacity constraints bind and additional demand translates mainly into inflation rather than higher employment.
- ▶ **Friedman's plucking model** (Friedman, 1964) mirrors this logic:
 - ▶ output typically fluctuates near a **ceiling** (capacity),
 - ▶ recessions “pluck” the economy below the ceiling; recoveries largely reflect the return toward it.
- ▶ **Policy emphasis differs:**
 - ▶ Friedman highlights monetary policy as the key tool to stabilize contractions and limit deep “plucks”.
 - ▶ Keynes places greater weight on fiscal policy.

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Departure from the Natural Rate

- ▶ Historically, the key departure from the ideal neoclassical world is the acknowledgment of information frictions leading to imperfect price adjustments.
- ▶ We consider the framework detailed in Section 3 and the optimization problem faced by unions under information frictions.
- ▶ We assume that unions do not have updated information on the state of the economy: one-period delayed information.

Information Frictions: Wage Setting with $t-1$ Information

- ▶ Unions maximize (12.5) with information from the previous period $t-1$, which implies the wage

$$W_t = \mu^w E_{t-1}\{P_t Y_t L_t^\eta\}$$

in a symmetric equilibrium.

- ▶ Labor demand is unchanged:

$$L_t = \frac{1}{\mu_t} \frac{P_t Y_t}{W_t}. \quad (12.9)$$

- ▶ ...but wages are set with the previous-period information.
- ▶ This framework generates a short-run Phillips curve relation around the natural rate.

Log-Linearization: Wage and Labor Equations

- A first order approximation around the non-stochastic steady state of the two above equations:

$$\hat{W}_t = E_{t-1} \{ \hat{P}_t + \eta \hat{L}_t + \hat{Y}_t \}, \quad (12.10)$$

$$\hat{L}_t = \hat{P}_t + \hat{Y}_t - \hat{W}_t - \hat{\mu}_t. \quad (12.11)$$

in which a variable with a hat denotes log-deviations with respect to the steady state.

- Take E_{t-1} of (12.11) and substitute into (12.10):

$$E_{t-1} \hat{L}_t = - \frac{E_{t-1} \{ \hat{\mu}_t \}}{1 + \eta}$$

and therefore $E_{t-1} \hat{L}_t = E_{t-1} \hat{L}_t^n$.

Expectations-Augmented Phillips Curve

- ▶ Combining (12.10) and (12.11) yields deviations of employment from the expected natural level:

$$\hat{L}_t - E_{t-1}\hat{L}_t^n = (\hat{P}_t + \hat{Y}_t) - E_{t-1}(\hat{P}_t + \hat{Y}_t) - (\hat{\mu}_t - E_{t-1}\hat{\mu}_t).$$

- ▶ **Interpretation:**

- ▶ Unexpected movements in nominal spending $(\hat{P}_t + \hat{Y}_t) - E_{t-1}(\hat{P}_t + \hat{Y}_t)$ raise employment relative to its expected natural level.
- ▶ Unexpected increases in markups $(\hat{\mu}_t - E_{t-1}\hat{\mu}_t)$ reduce employment relative to its expected natural level.
- ▶ The effects are symmetric: positive and negative surprises move employment in opposite directions.

Expectations-Augmented Phillips Curve

- Using $\hat{L}_t = \alpha_L^{-1}(\hat{Y}_t - \hat{A}_t)$, we obtain the expectations-augmented Phillips curve:

$$\pi_t = E_{t-1}\pi_t + k(\hat{Y}_t - \hat{Y}_t^n) + v_t, \quad (12.12)$$

where $\pi_t = \hat{P}_t - \hat{P}_{t-1} = p_t - p_{t-1}$ and $k \equiv \frac{1-\alpha_L}{\alpha_L}$.

- The composite shock term is

$$v_t = \frac{\eta + \alpha_L}{1 + \eta}(\hat{\mu}_t - E_{t-1}\hat{\mu}_t) - (\hat{A}_t - E_{t-1}\hat{A}_t),$$

and the (log) natural output is

$$\hat{Y}_t^n \equiv \hat{A}_t - \frac{\alpha_L}{1 + \eta}\hat{\mu}_t.$$

Phillips Curve with Unemployment

- Combine (12.7) and (12.9) to obtain

$$\frac{L_t}{L_t^s} = \mu_t^{\frac{1}{\eta}} L_t^{1+\frac{1}{\eta}}.$$

- Using $u_t = 1 - \frac{L_t}{L_t^s}$ and the approximation $u_t \approx -\ln(L_t/L_t^s)$ for small u :

$$u_t - u^n = -\left(1 + \frac{1}{\eta}\right)(\hat{L}_t - \hat{L}_t^n)$$

.

- Therefore, (12.12) can be written as:

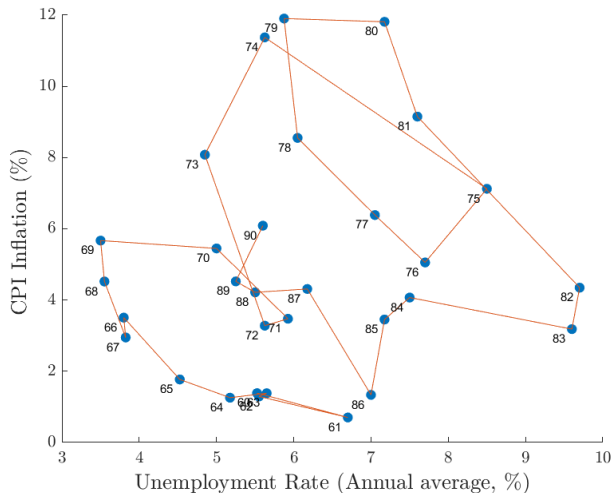
$$\pi_t = E_{t-1}\pi_t - k_u(u_t - u^n) + v_t, \quad (12.13)$$

$$\text{with } k_u = \frac{(1-\alpha_L)\eta}{1+\eta}.$$

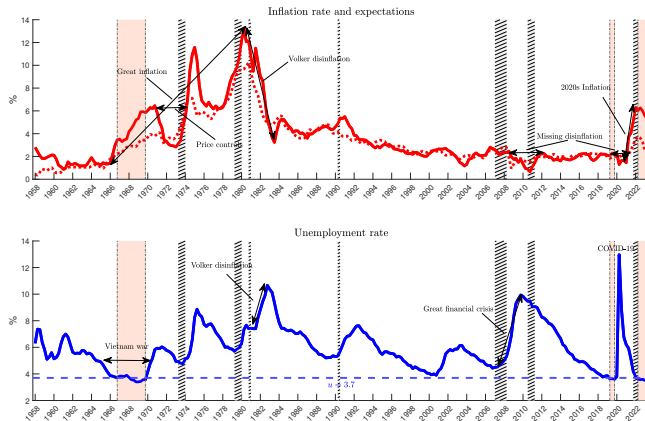
Key Features of the Expectations-Augmented Phillips Curve

- ▶ **Common foundations:** Output-gap and unemployment formulations measure slack differently, but share the same structure and policy implications for the inflation–activity trade-off.
- ▶ **Short-run non-neutrality:** Monetary disturbances have transitory real effects. Under rational expectations, adjustment can occur quickly (even within one period).
- ▶ **Only surprises matter:** Systematic policy cannot permanently raise real activity. Only unexpected inflation can move output or unemployment away from natural levels.
- ▶ **Symmetry around the natural rate:** Positive and negative deviations from the natural rate generate symmetric inflationary or disinflationary pressures.
- ▶ **Inflation is not pinned down:** The model does not uniquely determine the inflation rate. Real variables return to natural levels once expectations align with outcomes, not necessarily with the target.

Great Inflation: Inflation vs. Unemployment



Inflation and Unemployment in the US (Time Series)



- Evolution of U.S. inflation and inflation expectations (top panel) and the unemployment rate (bottom panel).

Great Inflation through the Phillips Curve Lens

- ▶ In the mid-1960s, inflation rose while unemployment was low, seemingly consistent with a short-run Phillips-curve trade-off.
- ▶ Monetary tightening in the late 1960s reduced output and raised unemployment—movement along the short-run curve.
- ▶ Over time, inflation expectations drifted upward, shifting the Phillips curve and weakening the apparent trade-off.
- ▶ Policy easing, wage and price controls, and adverse supply shocks contributed to stagflation: high inflation alongside high unemployment.
- ▶ The expectations-augmented Phillips curve explains why higher expected inflation translates into higher actual inflation at a given unemployment rate.
- ▶ The Volcker disinflation reduced inflation by tightening policy and lowering inflation expectations—at the cost of a sharp rise in unemployment.

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Persistent Monetary Non-Neutralities

- ▶ The expectations-augmented Phillips curve limits the inflation–output trade-off to the short run.
- ▶ Inflation expectations anchor the economy around the natural rate of output (or unemployment).
- ▶ Under rational expectations, one-time increases in nominal spending have only transitory real effects.
- ▶ Empirically, monetary shocks often generate hump-shaped responses in real activity.
- ▶ New-Keynesian models explain persistence through nominal rigidities and staggered price setting.

New-Keynesian Phillips Curve (NKPC)

The benchmark NKPC (from Chapter 5):

$$\pi_t - \pi = \kappa(\hat{Y}_t - \hat{Y}_t^n) + \beta E_t(\pi_{t+1} - \pi). \quad (12.14)$$

- ▶ The equation is similar to (12.12), but it features forward-looking inflation expectations $E_t(\pi_{t+1} - \pi)$ rather than past-period expectations of current inflation.
- ▶ This reflects the forward-looking behavior of price setters under Calvo pricing.
- ▶ As a result, monetary shocks can have persistent effects on output and inflation.
- ▶ Keeping inflation on target requires the entire expected path of output gaps to be zero, not only the current gap.

Nominal Spending and the Price-Level Phillips Curve

Log nominal spending follows:

$$\tilde{y}_t = \pi + \tilde{y}_{t-1} + \varepsilon_t. \quad (12.15)$$

- ▶ ε_t is white noise, π is the inflation target, and $\tilde{y}_t = p_t + y_t$.
- ▶ The NKPC can be rewritten as:

$$p_t - p_{t-1} - \pi = \kappa(\tilde{y}_t - p_t) + \beta E_t(p_{t+1} - p_t - \pi).$$

- ▶ This stochastic second-order difference equation can be written using the lag operator L (Chapter 7):

$$\left[L^{-1} - \left(1 + \frac{1}{\beta} + \frac{\kappa}{\beta} \right) + \frac{1}{\beta} L \right] p_t = -\frac{\kappa}{\beta} \tilde{y}_t - \frac{1 - \beta}{\beta} \pi.$$

Eigenvalues

- ▶ Characterizing the previous equation in terms of eigenvalues γ_1, γ_2 associated with the characteristic polynomial yields

$$[\mathbf{L}^{-1} - (\gamma_1 + \gamma_2) + \gamma_1\gamma_2\mathbf{L}] p_t = -\gamma_1\gamma_2\kappa\tilde{y}_t - \gamma_1\gamma_2(1 - \beta)\pi,$$

with $0 < \gamma_1 < 1 < \gamma_2$.

- ▶ The eigenvalues satisfy

$$\gamma_1 + \gamma_2 = 1 + \frac{1}{\beta} + \frac{\kappa}{\beta}, \quad (12.16)$$

$$\gamma_1\gamma_2 = \frac{1}{\beta}. \quad (12.17)$$

- ▶ Hence

$$(1 - \gamma_1\mathbf{L})(1 - \gamma_2^{-1}\mathbf{L}^{-1})p_t = \gamma_1\kappa\tilde{y}_t + \gamma_1(1 - \beta)\pi.$$

Solving for Prices

- Solving the previous equation (as in Chapter 7) gives

$$p_t = \gamma_1 p_{t-1} + \frac{\kappa}{\beta(\gamma_2 - 1)} \tilde{y}_t + \left(\frac{\kappa}{\beta(\gamma_2 - 1)^2} + \frac{1 - \beta}{\beta(\gamma_2 - 1)} \right) \pi.$$

- Using (12.16) and (12.18), this simplifies to

$$p_t = \gamma_1(\pi + p_{t-1}) + (1 - \gamma_1)\tilde{y}_t. \quad (12.18)$$

- Equivalently, using (12.15):

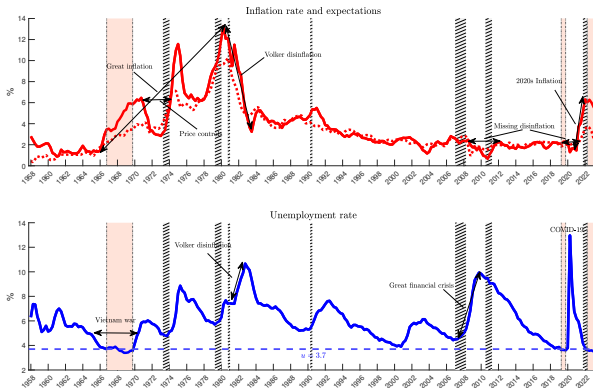
$$p_t = \pi + \gamma_1 p_{t-1} + (1 - \gamma_1)(\tilde{y}_{t-1} + \varepsilon_t).$$

Propagation of a Nominal Spending Shock

- ▶ A one-time innovation in nominal spending ε_t raises prices on impact by $(1 - \gamma_1)\varepsilon_t$.
- ▶ The shock propagates into future prices, decaying geometrically at rate γ_1 .
- ▶ Output responds on impact and then returns gradually toward its steady state at rate γ_1 .
- ▶ γ_1 governs the magnitude and persistence of real effects; it is inversely related to the slope κ of the NK Phillips curve.
- ▶ **Contrast with the New Classical Phillips curve:** with frictionless prices, only *unexpected* nominal shocks matter and real effects are much less persistent; NK frictions provide a mechanism for delayed and hump-shaped responses.
- ▶ Price rigidities and strategic complementarities amplify persistence and improve empirical fit.

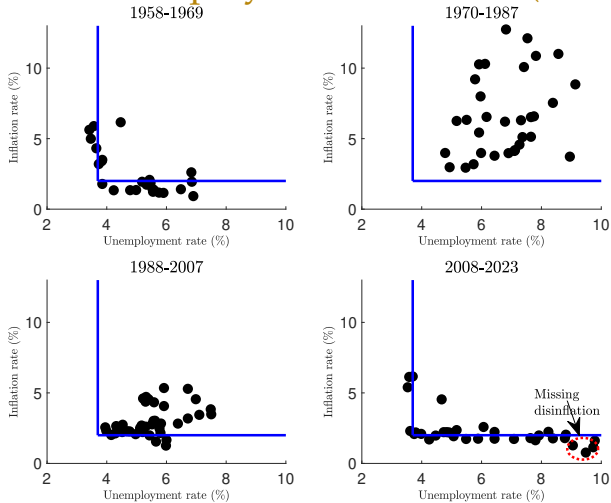
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Inflation and Unemployment in the US (Time Series)



- Evolution of U.S. inflation and inflation expectations (top panel) and the unemployment rate (bottom panel).

Inflation and Unemployment in the US (Scatter Plots)



- Scatter plots of CPI inflation and the unemployment rate for different U.S. samples.

Missing Disinflation and Post-COVID Inflation

- ▶ **Missing disinflation (Great Recession & COVID):** Slide 52 shows that large unemployment spikes did not generate large disinflations $\Rightarrow$ a very flat Phillips curve in slack markets.
- ▶ **Post-COVID surge:** Inflation rose sharply once labor markets became unusually tight (Slide 52) $\Rightarrow$ a steep/vertical segment near full employment.
- ▶ **Kinked “L-curve”** (Benigno & Eggertsson, 2023): scatter evidence in Slide 53; vertical segment around $u^n \approx 3.7\%$, flat segment around $\pi \approx 2\%$.

Slack vs. Tight Labor Markets

The model assumes the following **piecewise wage setting**:

$$W_t = \begin{cases} W_{t-1}(\Pi_t^e)^{\delta_\pi} & \text{if } u_t > u^n \\ W_t^{\text{flex}} = \mu^w P_t Y_t L_t^\eta & \text{if } u_t = u^n \end{cases}$$

- ▶ **Slack labor market** ($u_t > u^n$): wages do not fall below the previous level and adjust to expected inflation Π_t^e via $\delta_\pi \geq 0$.
- ▶ **Tight labor market** ($u_t = u^n$): wages follow the flexible-wage outcome (Section 3).

In log deviations from steady state, real wages are:

$$\hat{w}_t = \begin{cases} \hat{w}_{t-1} - (\pi_t - \pi) + \delta_\pi E_t(\pi_{t+1} - \pi) & \text{if } u_t > u^n \\ \hat{Y}_t + \eta \hat{L}_t & \text{if } u_t = u^n \end{cases} . \quad (12.19)$$

NKPC in Terms of Marginal Costs

- ▶ The NKPC can be written in terms of real marginal costs:

$$\pi_t - \pi = k_m(\hat{\mu}_t + \hat{w}_t + \hat{L}_t - \hat{Y}_t) + \beta E_t(\pi_{t+1} - \pi), \quad (12.20)$$

where $k_m > 0$.

- ▶ **Tight labor market** ($u_t = u^n$): substitute the second line of (12.19) into (12.20) to obtain

$$\pi_t - \pi = k_m[(1 + \eta)\hat{L}_t + \hat{\mu}_t] + \beta E_t(\pi_{t+1} - \pi), \quad (12.21)$$

- ▶ **Slack labor market** ($u_t > u^n$): using the first line of (12.19) yields

$$\pi_t - \pi = \tilde{k}_m[\hat{w}_{t-1} + (1 - \alpha_L)\hat{L}_t + \hat{\mu}_t - \hat{A}_t] + \tilde{k}_\pi E_t(\pi_{t+1} - \pi), \quad (12.22)$$

with $\tilde{k}_m = \frac{k_m}{1+k_m}$ and $\tilde{k}_\pi = \frac{\beta+k_m\delta_\pi}{1+k_m}$.

Key Implications

Key implications of (12.21) and (12.22):

- ▶ The Phillips curve is flatter in a slack labor market (since $\tilde{k}_m < k_m$).
- ▶ The response to a markup shock (e.g. an energy/oil-price shock) is smaller when the labor market is slack.
- ▶ Since (12.22) contains lagged real wages, inflation is more sluggish in slack labor markets.
- ▶ The sensitivity to inflation expectations, $\tilde{k}_\pi$, can be larger or smaller than β depending on the degree of wage indexation δ_π .

Implications for Unemployment

Tight labor market ($u_t = u^n$)

- ▶ With flexible wages, unemployment remains at the natural rate.
- ▶ From (12.21), inflation is determined by demand and supply shocks on a steep short-run Phillips curve.

Slack labor market ($u_t > u^n$)

- ▶ Using the log-linear approximation of labor-force participation (12.7):

$$\hat{L}_t^s = \frac{1}{\eta}(\hat{w}_t - \hat{Y}_t). \quad (12.23)$$

- ▶ Combining (12.23) and (12.20) implies the unemployment gap:

$$u_t - u^n = \hat{L}_t^s - \hat{L}_t = \frac{1}{\eta}\Omega_t - \frac{1}{\eta}\hat{\mu}_t - \left(1 + \frac{1}{\eta}\right)\hat{L}_t,$$

$$\text{where } \Omega_t \equiv \frac{\pi_t - \pi - \beta E_t(\pi_{t+1} - \pi)}{k_m}.$$

An L-shaped NK Phillips Curve in Unemployment Space

Slack labor market (cont.)

- Using the previous expressions,

$$u_t - u^n = \hat{L}_t^s - \hat{L}_t = \frac{1}{\eta}\Omega_t - \frac{1}{\eta}\hat{\mu}_t - \left(1 + \frac{1}{\eta}\right)\hat{L}_t.$$

- Solving for employment as a function of the unemployment gap:

$$\hat{L}_t = -\frac{\eta}{1 + \eta}(u_t - u^n) + \frac{1}{1 + \eta}(\Omega_t - \hat{\mu}_t).$$

- Substituting into (12.22) yields:

$$\pi_t - \pi = \tilde{k}_u(\hat{w}_{t-1} - \gamma_u(u_t - u^n) + \gamma_\mu\hat{\mu}_t - \hat{A}_t) + \tilde{k}'_\pi E_t(\pi_{t+1} - \pi), \quad (12.24)$$

valid when $u_t > u^n$, for positive parameters $\tilde{k}_u, \tilde{k}'_\pi, \gamma_u, \gamma_\mu$.

Flat Phillips Curve Implications

- ▶ The key inequalities $\tilde{k}_u < \tilde{k}_m$ and $\gamma_u < 1$ imply a relatively flat inflation–unemployment trade-off in slack labor markets.
- ▶ Once unemployment returns to the natural rate, the Phillips curve is vertical with respect to unemployment rate and inflation is then driven by excess output/employment above their natural levels.
- ▶ This helps interpret the U.S. evidence: long periods of slack (flat segment) and sharp inflation pressures when the economy approaches full employment (see Slides 52 and 53).

Policy Implications

- ▶ The 1970s inflation occurred despite slack labor markets; it reflected adverse supply shocks and drifting inflation expectations, with wage indexation amplifying persistence.
- ▶ By contrast, the inflation surge of the 2020s emerged under unusually tight labor-market conditions, where demand stimulus and supply shocks interacted with a steeper aggregate-supply relationship.
- ▶ Interpreting events through a steep rather than flat Phillips curve changes policy trade-offs: disinflation may require relatively small unemployment increases when the economy operates near the steep segment, but larger unemployment increases once the economy operates in the slack regime.

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Main Takeaways

- ▶ The Phillips curve is an empirical relationship in search of a theory.
- ▶ In a flexible-wage benchmark, unemployment is pinned down by labor-market fundamentals (**natural rate**) and is **independent of monetary policy** in the long run.
- ▶ With information frictions, the short-run trade-off arises because only **surprises** in nominal spending/inflation move output and unemployment away from their natural levels.
- ▶ In New-Keynesian models, forward-looking price setting implies that monetary policy shocks can have persistent real effects.
- ▶ Combining wage rigidity with the NKPC yields an **L-shaped** inflation–unemployment relationship: flat in slack markets and steep near full employment—helping reconcile missing disinflation with the post-COVID inflation surge.

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